

Claude Certified Architect – Foundations (CCAR-F)

Bootcamp Syllabus · 4 Days · 4 Hours/Day · 16 Total Hours

Course Description

This intensive 4-day bootcamp prepares candidates to sit the Claude Certified Architect – Foundations (CCAR-F) exam. The program is built directly against the official exam blueprint and its five weighted domains, using scenario-based labs rather than lecture-heavy delivery — mirroring the format of the exam itself (60 scenario-based questions, 120 minutes, scaled passing score of 720/1000).

Audience & Prerequisites

- Solution architects, AI engineers, and technical consultants designing or implementing production Claude applications
- Anthropic lists no formal prerequisite; 6+ months of hands-on Claude experience (API, Claude Code, or Agent SDK) is strongly recommended
- Pre-work: complete Anthropic Academy's “Building with the Claude API,” “Claude Code 101,” and “Introduction to MCP” (free, self-paced) before Day 1

Learning Objectives

By the end of this bootcamp, participants will be able to:

- Design and evaluate agentic and multi-agent architectures, including orchestrator-worker and evaluator-optimizer patterns
- Configure Claude Code workflows using CLAUDE.md, permission models, and CI/CD integration
- Design tool descriptions and MCP integrations that support reliable tool selection
- Apply structured prompting techniques (few-shot, chain-of-thought, JSON-schema output, validation-retry) to production use cases
- Manage context windows, caching, and token budgets for reliability at scale
- Complete a full-length, timed practice exam under realistic conditions

Schedule Overview

Day	Domain Focus	Key Topics	Exam Weight
1	Domain 1: Agentic Architecture & Orchestration	Agent loops, stop_reason, multi-agent topologies, subagent isolation, task decomposition	27%
2	Domain 3: Claude Code Configuration & Workflows + Domain 2 (intro): Tool Design & MCP	CLAUDE.md hierarchy, permission models, headless CI/CD; tool description design basics	20% / 18%
3	Domain 2 (cont.): MCP Integration + Domain 4: Prompt Engineering & Structured Output	MCP end to end, tool_choice, structured errors; few-shot, CoT, JSON-schema tool use, validation-retry loops	18% / 20%
4	Domain 5: Context Management & Reliability + Integration Lab + Mock Exam	Lost-in-the-middle, prompt caching, token budgets; capstone scenario lab; timed practice exam & review	15%

Day 1 — Agentic Architecture & Orchestration (Domain 1, 27%)

Time	Activity	Format
0:00–0:20	Program orientation; exam format walkthrough; pre-work check	Discussion
0:20–1:30	Agent loop fundamentals: autonomy, tool use, action loop, stop_reason	Lecture + demo
1:30–1:40	Break	
1:40–2:40	Multi-agent topologies: hub-and-spoke, orchestrator-worker, evaluator-optimizer, subagent isolation	Lecture + case study
2:40–3:40	Lab: decompose a real client workflow into a multi-agent design	Hands-on lab
3:40–4:00	Debrief and Q&A	Discussion

Day 2 — Claude Code Workflows (Domain 3, 20%) & MCP Introduction (Domain 2, 18%)

Time	Activity	Format
0:00–0:15	Recap Day 1; open questions	Discussion
0:15–1:15	CLAUDE.md hierarchy, slash commands, Skills, permission models	Lecture + demo
1:15–2:15	Headless Claude Code in CI/CD: automated review, test generation, PR workflows	Lecture + demo
2:15–2:25	Break	
2:25–3:30	Tool design fundamentals: tool descriptions as the selection mechanism, tool_choice	Lecture + lab
3:30–4:00	Lab: write and test a tool description for a client scenario	Hands-on lab

Day 3 — MCP Integration (Domain 2, 18%) & Prompt Engineering (Domain 4, 20%)

Time	Activity	Format
0:00–0:15	Recap Day 2; open questions	Discussion
0:15–1:15	Model Context Protocol end to end: tools, resources, prompts; structured error handling	Lecture + demo
1:15–2:15	Few-shot and chain-of-thought techniques for reliability	Lecture + lab
2:15–2:25	Break	
2:25–3:30	JSON-schema tool use, structured output, and validation-retry loops	Lecture + lab
3:30–4:00	Lab: build a validated structured-output pipeline	Hands-on lab

Day 4 — Context Management & Reliability (Domain 5, 15%), Integration Lab, Mock Exam

Time	Activity	Format
0:00–0:15	Recap Day 3; open questions	Discussion
0:15–1:15	Context management: lost-in-the-middle, scratchpads, provenance, prompt caching, token budgets	Lecture + demo
1:15–2:15	Capstone lab: full architecture design integrating Domains 1–5 for a single client scenario	Hands-on lab
2:15–2:25	Break	
2:25–3:30	Timed mock exam: 30 scenario-based questions (half-length practice set)	Assessment
3:30–4:00	Mock exam review, domain-by-domain score breakdown, individual study plans for remaining gaps	Discussion

Materials & Resources

- Official Claude Certified Architect – Foundations Exam Guide (v1.0, effective July 2026)
- Anthropic Academy free courses: Building with the Claude API, Claude Code 101, Introduction to MCP
- Daily lab environment: Claude Code + Claude API sandbox access
- Slide decks and lab guides distributed per day via LMS
- Full-length (60-question, 120-minute) take-home practice exam, to be completed before scheduling the official exam

Assessment & Completion Criteria

- Attendance and active participation across all 4 sessions
- Completion of all daily hands-on labs
- Day 4 mock exam attempted (no minimum score required to complete the bootcamp; used as a diagnostic)
- Recommended: score 80%+ on the full-length take-home practice exam before booking the official Pearson VUE exam

Notes for Facilitators

- All exam questions are scenario-based — avoid pure lecture; keep lecture segments under 60 minutes before returning to a lab or case discussion
- Domain 1 (27%) and the combined Domain 2/4 blocks carry the most exam weight — protect their lab time if the schedule slips
- Exam guide is versioned (v1.0 as of July 2026) — confirm the current blueprint before each cohort, as domain weights can shift between versions